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**, Mastery in Data Analytics and Visualizations and Career Advancement ,**  
**Labsheet-7**

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

df = pd.read_csv("/Mall_Customers.csv")

df.head()

    CustomerID Gender Age Annual Income (k$) Spending Score (1-100)
0 1 Male 19 15 39
1 2 Male 21 15 81
2 3 Female 20 16 6
3 4 Female 23 16 77
4 5 Female 31 17 40

print("Shape of dataset:", df.shape)

print("\nDataset Information:")
df.info()

print("\nMissing Values:")
print(df.isnull().sum())

Shape of dataset: (200, 5)

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 # Column Non-Null Count Dtype
---  ---
0 CustomerID 200 non-null int64
1 Gender 200 non-null object
2 Age 200 non-null int64
3 Annual Income (k$) 200 non-null int64
4 Spending Score (1-100) 200 non-null int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB

Missing Values:
CustomerID 0
Gender 0
Age 0
Annual Income (k$) 0
Spending Score (1-100) 0
dtype: int64

df = df.drop_duplicates()
print("Shape after removing duplicates:", df.shape)

Shape after removing duplicates: (200, 5)

X = df[["Age", "Annual Income (k$)", "Spending Score (1-100)"]]

X.head()
```

```
0 19 15 39
1 21 15 81
2 20 16 6
3 23 16 77
4 31 17 40
```

```
scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

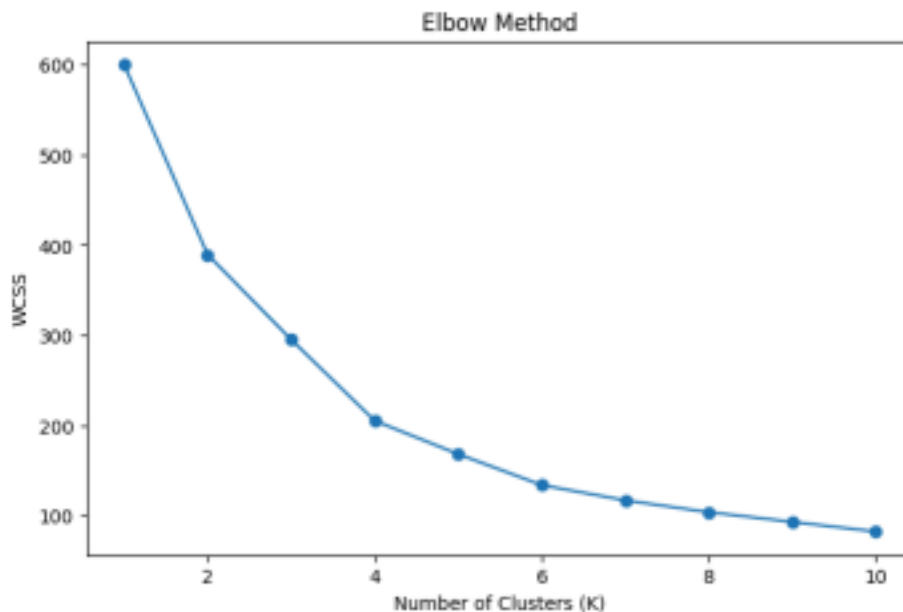
print("Scaled data:")
print(X_scaled[:5])

Scaled data:
[[-1.42456879 -1.73899919 -0.43480148]
 [-1.28103541 -1.73899919  1.19570407]
 [-1.3528021  -1.70082976 -1.71591298]
 [-1.13750203 -1.70082976  1.04041783]
 [-0.56336851 -1.66266033 -0.39597992]]
```

```
wcss = []

for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

# Plot Elbow graph
plt.figure(figsize=(8, 5))
plt.plot(range(1, 11), wcss, marker="o")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()
```



```
silhouette_scores = []

for k in range(2, 11):
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    labels = kmeans.fit_predict(X_scaled)

    score = silhouette_score(X_scaled, labels)
    silhouette_scores.append(score)

# Display scores
for k, score in zip(range(2, 11), silhouette_scores):
    print(f"K = {k}, Silhouette Score = {score:.3f}")
```

```

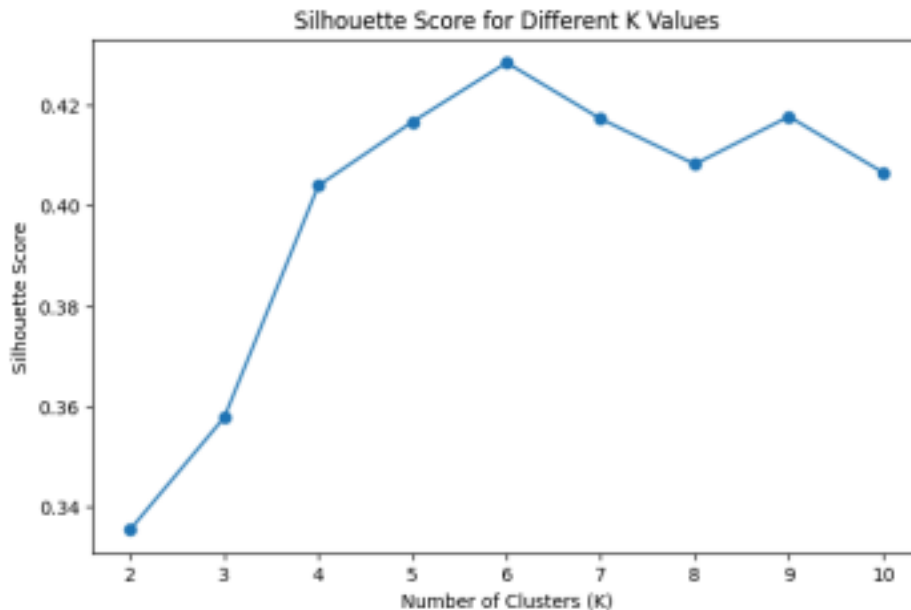
K = 2, Silhouette Score = 0.335
K = 3, Silhouette Score = 0.358
K = 4, Silhouette Score = 0.404
K = 5, Silhouette Score = 0.417
K = 6, Silhouette Score = 0.428
K = 7, Silhouette Score = 0.417
K = 8, Silhouette Score = 0.408
K = 9, Silhouette Score = 0.418
K = 10, Silhouette Score = 0.407

```

```

plt.figure(figsize=(8, 5))
plt.plot(range(2, 11), silhouette_scores, marker="o")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("Silhouette Score")
plt.title("Silhouette Score for Different K Values")
plt.show()

```



```
kmeans = KMeans(n_clusters=5, random_state=42, n_init=10)
```

```
df["Cluster"] = kmeans.fit_predict(X_scaled)
```

```
df.head()
```

```

CustomerID Gender Age Annual Income (k$) Spending Score (1-100) Cluster
0 1 Male 19 15 39 1 1 2 Male 21 15 81 1 2 3 Female 20 16 6 0 3 4 Female
23 16 77 1 4 5 Female 31 17 40 1

```

Next steps: [Generate code with df](#)  
[New interactive sheet](#)

```

centers = scaler.inverse_transform(kmeans.cluster_centers_)
centers_df = pd.DataFrame(
    centers,
    columns=["Age", "Annual Income (k$)", "Spending Score (1-100)"]
)
centers_df

```

Age Annual Income (k\$) Spending Score (1-100)

```
0 46.250000 26.750000 18.350000
1 25.185185 41.092593 62.240741
2 32.875000 86.100000 81.525000
3 39.871795 86.102564 19.358974
4 55.638298 54.382979 48.851064
```

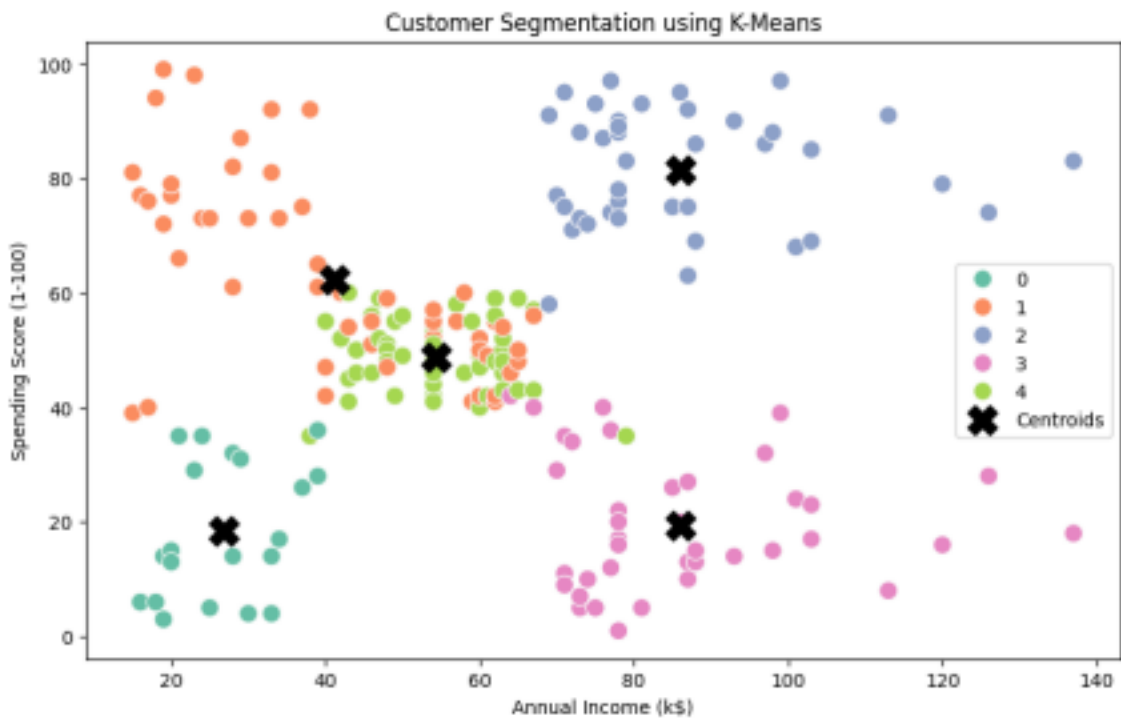
Next steps: [Generate code with centers\\_df](#) [New interactive sheet](#)

```
plt.figure(figsize=(10, 6))

sns.scatterplot(
    data=df,
    x="Annual Income (k$)",
    y="Spending Score (1-100)",
    hue="Cluster",
    palette="Set2",
    s=100
)

plt.scatter(
    centers[:, 1],
    centers[:, 2],
    marker="X",
    s=250,
    c="black",
    label="Centroids"
)

plt.title("Customer Segmentation using K-Means")
plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score (1-100)")
plt.legend()
plt.show()
```



```
cluster_analysis = df.groupby("Cluster")[
    ["Age", "Annual Income (k$)", "Spending Score (1-100)"]
].mean()

cluster_analysis
```

Age Annual Income (k\$) Spending Score (1-100)

Cluster

```
0 46.250000 26.750000 18.350000
1 25.185185 41.092593 62.240741
2 32.875000 86.100000 81.525000
3 39.871795 86.102564 19.358974
4 55.638298 54.382979 48.851064
```

Next steps: [Generate code with cluster\\_analysis](#) [New interactive sheet](#)

```
cluster_counts = df["Cluster"].value_counts().sort_index()

print("Number of customers in each cluster:")
print(cluster_counts)

Number of customers in each cluster:
Cluster
0 20
1 54
2 40
3 39
4 47
Name: count, dtype: int64

print("Customer Segmentation using K-Means")
print("-----")

print("Number of clusters:", 5)
print("Silhouette Score:", silhouette_score(X_scaled, df["Cluster"]))

print("\nCluster-wise Average:")
print(cluster_analysis)

Customer Segmentation using K-Means
-----
Number of clusters: 5
Silhouette Score: 0.41664341513732767

Cluster-wise Average:
Age Annual Income (k$) Spending Score (1-100)
Cluster
0 46.250000 26.750000 18.350000
1 25.185185 41.092593 62.240741
```

# Theory Questions

## 1. What is clustering? How does it differ from classification?

**Clustering** is an unsupervised machine learning technique used to divide data into groups called clusters. Data points within the same cluster are more similar to each other than to points in other clusters.

**Classification** is a supervised learning technique in which data is assigned to predefined classes using labeled training data.

**Difference:** Clustering uses **unlabeled data**, while classification uses **labeled data**.

## 2. Explain the working principle of the K-Means Clustering algorithm.

K-Means is a clustering algorithm that divides data into a fixed number of **K clusters**.

The main steps are:

1. Select the number of clusters K.
2. Initialize K centroids randomly.
3. Assign each data point to its nearest centroid.
4. Calculate the mean of all points in each cluster to obtain new centroids.
5. Repeat the assignment and centroid calculation steps.
6. Stop when the centroids become stable.

Thus, K-Means forms groups based on the distance between data points and cluster centroids.

## 3. Why is feature scaling important before applying K-Means clustering?

K-Means uses **distance calculations** to assign data points to clusters. If features have different scales, a feature with larger numerical values can have more influence on the distance.

For example, income may have values in thousands while age may have values between 18 and 70. Without scaling, income could dominate the clustering process.

Therefore, **standardization or normalization** is used so that all features contribute more equally and meaningful clusters can be obtained.

## 4. What is the Elbow Method, and how does it help determine the optimal number of clusters?

The **Elbow Method** is used to determine a suitable value of K in K-Means clustering.

It calculates **Within-Cluster Sum of Squares (WCSS)** for different values of K. As K increases, WCSS decreases because the clusters become smaller.

A graph of K against WCSS is plotted. The point where the reduction in WCSS starts becoming slower forms an **elbow**. This point is generally selected as the optimal number of clusters.

## 5. What is the Silhouette Score? How is it used to evaluate clustering

## performance?

The **Silhouette Score** is a metric used to measure the quality of clustering. It checks how similar a data point is to its own cluster compared with other clusters.

Its value generally ranges from **-1 to +1**.

- A value close to **+1** indicates well-separated clusters.
- A value around **0** indicates overlapping clusters.
- A negative value indicates that some points may be assigned to the wrong cluster.

Therefore, a higher Silhouette Score generally represents better clustering performance.

## 6. Why is K-Means considered an unsupervised machine learning algorithm?

K-Means is considered an **unsupervised machine learning algorithm** because it does not require labeled or predefined output data.

It receives only the input features and identifies natural groups based on similarities and distances between data points. The algorithm itself determines which data points belong to which clusters.

For example, in customer segmentation, customers can be grouped according to **age, income, and spending score** without having predefined customer categories.

## 7. Mention any four real-world applications of customer segmentation.

Customer segmentation has many applications in business. Four important applications are:

1. **Targeted marketing:** Businesses can create different campaigns for different customer groups.
2. **Personalized recommendations:** Products can be recommended according to customer interests.
3. **Customer retention:** Special offers can be provided to customers who may stop using a service.
4. **Customer engagement:** Businesses can design suitable strategies to improve interaction with different customer groups.

Customer segmentation helps organizations improve marketing and overall business performance.

## 8. What are the limitations of the K-Means algorithm?

K-Means has several limitations:

- The number of clusters **K must be selected in advance**.
- Results can depend on the **initial selection of centroids**.
- It is sensitive to **outliers and noisy data**.
- It works best when clusters have relatively simple and similar shapes.
- It may not perform well when clusters have very different sizes or densities.

Therefore, proper preprocessing and selection of K are important for obtaining good results.

## 9. How can businesses use customer segmentation to improve marketing and customer retention?

Businesses can divide customers into groups based on characteristics such as **income, spending behavior,**

**and purchasing patterns.** Different marketing strategies can then be created for each group.

For example, high-spending customers can receive loyalty rewards, while customers with low engagement can receive discounts or personalized offers.

This helps businesses provide more relevant services, increase customer satisfaction, improve engagement, and encourage customers to remain loyal.

## 10. Compare K-Means Clustering with Hierarchical Clustering.

| K-Means                                                                                                                                                                                              | Hierarchical Clustering                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Divides data into a fixed number of clusters.                                                                                                                                                        | Builds a hierarchy of clusters.                        |
| Uses centroids to form clusters.                                                                                                                                                                     | Uses a tree-like structure called a dendrogram.        |
| Number of clusters K is selected beforehand.                                                                                                                                                         | Clusters can be selected by cutting the dendrogram.    |
| Generally faster for large datasets.                                                                                                                                                                 | Can be slower for large datasets.                      |
| Suitable for customer segmentation and large datasets.                                                                                                                                               | Useful for understanding relationships between groups. |
| <b>In short</b> , K-Means is useful when the required number of clusters is known, whereas Hierarchical Clustering is useful when we want to study the hierarchical relationships among data points. |                                                        |